

Article

Board Characteristics and Corporate Sustainability Reporting: Evidence from Chinese Listed Companies

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Abstract: This paper investigates the impact of board characteristics (such as board size, board independence, CEO Duality, board meetings, and committee) on corporate sustainability reporting (CSR) in China. Unlike previous studies, we examine this relationship in light of the three pillars of corporate sustainability reporting: economic, environmental, and social. Using both the Fully Modified Ordinary Least Square (FMOLS) and Dynamics Ordinary Least Square (DOLS) empirical models, our data, which comprises 9842 firm-year observations from both the Shanghai and Shenzhen stock exchanges covering the 2006–2019 fiscal periods, reveals that board independence, a larger board size, and a board sustainability committee promote CSR in the three CSR pillars. However, the study shows that CEO duality is more likely to impede CSR. While the current study seems to provide some understanding regarding the relationship between board characteristics and corporate sustainability reporting to corporate boards, regulators, and practitioners interested in advancing the course of CSR, some significant knowledge gaps still need to be explored in future studies. Future research may consider other board characteristics as well as explore other CSR indices like integrated reporting and triple-bottom-line approaches.

Keywords: corporate sustainability reporting; economic sustainability reporting; environmental sustainability reporting; social sustainability reporting; board characteristics; Chinese companies



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1. Introduction

With the recent global concerns about achieving the United Nations 2030 sustainable development agenda, firms worldwide have been under increasing pressure to engage in environmental, social, and economic sustainable practices [1]. Chinese businesses, especially the manufacturing sector, are responsible for about 70% of total emissions and are therefore under significant pressure to foster economic growth that benefits the environment and society [2]. Chinese industries are expected to do more to protect the environment by fighting climate change [3]. These companies have additional responsibilities to protect society, including providing safe and healthy workplaces and assisting the disabled, chronically ill, or elderly [4].

Based on the above, many Chinese businesses, especially those in the manufacturing sector, have been forced to rethink their investment strategies and, over time, adopt environmentally and socially responsible practices [5]. More importantly, stakeholders have recently asked for stricter corporate control mechanisms that put more emphasis on corporate sustainability reporting [6] (CSR). CSR shows how businesses promote sustainable development [7]. These three areas, or pillars—economic, environmental, and social—are essential for sustainable development. Together, they make up the “trifecta” of CSR. The economic pillar concerns ethical and transparent accounting practices and regulatory compliance. Environmental sustainability concerns how companies lower their

carbon footprints by reducing waste and pollution and improving air quality [8]. Social sustainability reports on the effects of businesses on the social systems in which they operate. It includes ensuring healthy labour practices, providing decent work, protecting human rights, protecting disadvantaged customers, and providing special services to the elderly, disabled, and chronically ill [9]. Adherence to environmental, economic, and social sustainability reporting indicates firms' effort to enhance societal well-being and project a good image of the firms to attract more equity capital [10]. Thus, scholars, the government, and policymakers are becoming more interested in improving CSR.

Previous research has identified board characteristics as an essential corporate governance mechanism with the potential to improve CSR [1,11–17]. For instance, according to Shahbaz et al. (2020) [18], the board is the highest management level in a company and, therefore, responsible for making numerous significant decisions, such as reporting on corporate sustainability to various stakeholder groups. Therefore, empirically, Jizi (2017) [19] shows that board management supports social sustainability reporting among FTSE 350 firms. In another study, Husted & de Sousa-Filho (2019) [13] argued that board independence encourages environmental, social, and governance (ESG) disclosure in 176 Latin American publicly traded companies across different countries. Supporting these studies is the agency theory, which suggests that boards with high independence can closely monitor how businesses are operated and influence businesses to care more about what various stakeholders want and need [20]. The resource dependence theory also proposes that independent and diverse boards can improve a firm's overall social performance [21]. Aliyu (2019) [22] studied 24 non-public listed companies in Nigeria and established that companies with more independent board members are more likely to promote corporate environmental reporting. Endrikat et al. (2021) [23] also demonstrate that board independence promotes corporate social disclosures in 82 empirical studies. Similarly, Masud et al. (2018) [16] show that board characteristics affect environmental sustainability reporting in 88 organizations in South Asian countries.

On the contrary, other studies have reported that board characteristics may not essentially promote CSR. For instance, Mahmood et al. (2018) [15] studied the top 100 firms in Pakistan and observed that board independence may have nothing to do with whether the firm reports on sustainability or not. Husted and de Sousa-Filho (2019) [13] found that firms whose CEO, besides running the organization at the highest level, also holds the position of Chairman of the Board tend to abuse power and position and therefore undermine the importance of CSR. Similarly, Liu et al., 2020 [24] demonstrates that independent directors impede sustainable operations in Chinese firms. A study conducted by Hu and Loh (2018) [12] in Singapore shows that CEO duality does not affect sustainability disclosures in 462 Singaporean firms. Akbas (2016) [25] observed that board independence in Turkish non-financial companies is unrelated to environmental disclosure. Additionally, the study by Aliyu (2019) [22] concludes that a larger board size is insignificantly related to corporate environmental reporting in Nigerian non-public listed companies.

Given the inconclusive nature of the above empirical studies, at this point, it is difficult to determine a priori which effect is dominant, so thorough examination of the relationship between corporate governance and sustainability reporting is needed. Possible reasons for these contradictory findings may have emanated from the study context, datasets and measurement for corporate sustainability reporting. Furthermore, the majority of these studies focus on firms in developed countries such as the United States and Europe ([13,17,18,26]; inter-alia). Only few studies attempt to link board characteristics and CSR from the perspective of the developing or emerging economies like China. Given that China's industrial sector has been at the forefront of climate change debates for decades, and as one of the world's largest emitter of GHG, China is an interesting case for this study as we pay special attention to the manufacturing industries in China. According to Qunhui (2021) [27], China has been the world's largest manufacturing hub with over 220 types of industrial products listed in the UN industrial classification [28]. Given this, the activities of these companies have significant social, economic, and environmental implications. In

its quest to enforce sustainability regulations enshrined in Chinese company laws, the Chinese government has mandated all listed companies to contribute to the sustainable development agenda. Therefore, the Environmental Protection Administration announced the disclosure of Environmental Information in 2003, followed by the Shenzhen Stock Exchange (SZSE) and Shanghai Stock Exchange (SSE), and promulgated laws in 2006 and 2008, respectively, for listed companies to disclose social and environmental information. Similarly, in 2008, the State-Owned Assets Supervision and Administration Commission (SASAC) of the State Council also released “Guiding Advice on Filling Social Responsibility by Central Enterprises” documents to encourage them to show their extemporaneous role in fulfilling CSR. As a result, many industries have implemented some internal strategies and initiatives throughout their management process, placing greater emphasis on elevating transparency and enhancing communication with the public and stakeholders through non-financial disclosure while integrating the concept of sustainability into their core business strategy [29]. Considering these firm-level initiatives, it would be interesting to see whether board characteristics influence corporate sustainability reporting in China.

Nevertheless, in this study, we depart a little from the previous literature by looking at the problem from a wider spectrum. Specifically, we attempt to investigate the influence of board characteristics on CSR in the light of the three CSR pillars—social, economic, and environmental. Previous studies examined the impact of board characteristics on corporate social responsibility [5,19,23,30–32], environmental sustainability [10,16,22], or firm performance [33–36]. In other words, these studies have focused on one or two pillars of CSR, primarily economic or social sustainability reporting [5,16,24,31,32]. Considering that all the three areas, or pillars of CSR—economic, environmental, and social—are essential for sustainable development, focusing on one or two and ignoring the rest may be problematic and lead to erroneous conclusions about how board characteristics and CSR are linked. Furthermore, there may be evidence of a bi-directional causal relationship between board characteristics and CSR, leading to estimation issues or vulnerability to the reverse causality problem.

Motivated by the above, this study seeks to answer the following question: How does corporate governance, specifically the board characteristics, improve corporate sustainability reporting in Chinese firms? In answering this question, our research seeks to explore the effects of board characteristics on corporate sustainability reporting (CSR) under three key scopes: economic, environmental, and social reporting. Therefore, following Orazalin and Mahmood (2018) [37], we used the Global Reporting Initiative (GRI 3.1) guidelines published in 2011 to derive economic, environmental, and social sustainability reporting scores to create a composite index.

This research contributes to the literature in the following ways. For example, this research is one of the few that examine the effects of internal factors on CSR in the Chinese industry. Second, in this study, we address empirical challenges that may affect the causal influence of CSR on board characteristics. Thus, by creating a composite index of environmental, social, and economic sustainability reporting, we address the omitted factors that may limit the causal ability of CSR on board characteristics. Finally, while previous studies have only focused on shorter periods, mostly less than ten years, the current study investigates these issues using panel data covering 14 fiscal years.

The outcomes of the analysis suggest that while larger boards, board independence, board meetings, and board committee promote the disclosure of sustainability information, CEO duality impedes it. Based on these findings, our study offers the following suggestions. First, because our findings imply that a bigger board size encourages the disclosure of information about business sustainability, firms should consider a board size ranging from 3 to 20 members. Nonetheless, future research on the optimal board size would be interesting. Second, we propose increasing the number of independent directors on boards to ensure effective advocacy for stakeholder-driven concerns. Third, the board chairmanship and the position of the chief executive officer should be separated. In other words, both positions should not be held by one person. Fourth, the directors are also encouraged to attend all

scheduled board meetings to enhance the board's diligence in performing its duties. Lastly, we encourage the formation of a board committee devoted to sustainability performance and information reporting to help companies pursue economic, environmental and social performance and closely impact related goals.

The remainder of this study is organized as follows. The next section reports on contextual and institutional background of the study, the theoretical framework, literature, and the hypotheses. In the next section, we describe the research methodology. The next section is the presentation and discussion of the results. Finally, the last section concludes the study.

2. Contextual and Institutional Background of the Study

Corporate Sustainability Reporting (CSR) in China

Despite the numerous advantages of corporate sustainability reporting, such as increased efficiency, a reduction in social, environmental, and economic risks, and increased appreciation of the link between financial and non-financial performance, sustainability reporting is still a voluntary option for businesses in most countries around the world [38]. However, the situation is different for most Chinese companies. Corporate sustainability reporting has become necessary and compulsory for most companies in China [39].

After more than 30 years, China has achieved the fame of extraordinary and enviable economic growth to become the world's second-largest economy after the USA and a manufacturing hub for the global society [39]. High environmental issues and social costs still accompany the country's economic growth [40]. These adverse environmental and social impacts continue to pose challenges for the country and have attracted global concern [41]. Therefore, in response to these predicaments and concerns, China has sought to implement several regulatory policies and initiatives, including the innovative circular economy (C.E.), low-carbon economy policies, and a series of sustainability programmes, especially in environmental sustainability-oriented policies since the 1980s at the macro and micro levels [42]. These regulatory policies for environmental protection include "the 1989 Environmental Protection Law of the People's Republic of China", the 1992 China's Agenda 21, National Environmental Protection Agency, 1994a. Furthermore, "the 2003 Law of promoting cleaner production, Law of Safety production, the law on Desert Prevention and Transformation, the Law on Environment Impact Assessment, the Law on Radioactive Pollution Prevention and control", "the 2004 Law on Administrative Permission and the law on the Promotion of Renewable Energy in 2005", "the 2009 announcement by the State Council setting a countrywide goal for the reduction of carbon dioxide emission per unit GDP in 2020 by 40–45 per cent relative to the emissions in 2005 and reach the carbon peak by 2030, and becoming carbon neutral by 2060" [42]. The Environmental Protection Law of the people's Republic of China was revised and enacted on 1 January 2015. In June 2022, China's standard guidelines for enterprise ESG disclosure also took effect. All these provisions are available on the reporting exchange and will be updated as corporate reporting continues to develop in China. So far, China has shown that it is committed to eradicating this environmental menace through these regulatory policies [2].

Social issues began without any specific or emphatic legislative promulgation. However, they were captured in various pieces of legislation, such as corporate law, labour law, women's rights protection law, environmental law, and consumer protection laws [42]. Since most definitions did not suit corporate social responsibility principles, these practices were meant to go beyond regulatory compliance. The agitation for a comprehensive corporate social responsibility standard began in 2004 when corporate sustainability reporting standards and the core labour standard became rampant in China. Following these agitations, the Company Law of the People's Republic of China was first proposed to be amended to cater for corporate social responsibility in 2006. It coincides with the Chinese Communist Party's sixth plenary session, where the proposal was to build a harmonious society with a key focus on improving the social responsibility of businesses and organizations.

Therefore, after the proposal, the Shenzhen Stock Exchange (SZSE) made an effort in 2006 to put out CSR guidelines for companies in the Shenzhen (S.Z.) 100 index that was listed on the SZSE [42]. Subsequently, the Shanghai Stock Exchange (SSE) also followed suit, asking public firms, namely those in the corporate governance index, dual-listed firms, and firms in the financial industry, to disclose their CSR reports [43]. Since then, companies in the above categories have reported on corporate sustainability [44]. Similarly, in 2008, the State-Owned Assets Supervision and Administration Commission (SASAC) of the State Council released “*Guiding Advice on Filling Social Responsibility by Central Enterprises*”, urging corporations to show their role in fulfilling CSR.

Even though the government is committed to putting these policies and interventions into place, China still has a lot of environmental, economic, and social problems. As a result, China is criticized for being a major polluter and continues to lead other countries in carbon dioxide emissions [45]. This issue, combined with hazardous working conditions, occupational injuries, diseases in the Chinese mining industries, and labour-intensive manufacturing companies, casts doubt on China’s sustainability drive. Both Chinese and international news outlets have talked about these issues [41]. To deal with these problems, the Chinese government and policymakers have put pressure on businesses to adopt and report on sustainable practices.

However, to help provide a panacea to these issues arising from the Chinese business environment and enhance the issue of corporate sustainability reporting, corporate governance rules have been put in place to check these anomalies. The corporate governance code for the Chinese listed firms are as follows: (i) the number of directors should range from 3 to 13 for limited liability companies and 5 to 19 for listed companies; (ii) the directors and managers are answerable to the shareholders and should protect the company’s interest while performing their duties faithfully; (iii) a director’s appointment must not exceed three years unless it is subjected to reappointment for a further term; (iv) companies must ensure that the CEO and the chairmanship position must be separated. The CSRC also requires that listed company boards should comprise at least one-third of independent directors. Finally, in terms of directorship positions, one director cannot simultaneously hold more than five directorship positions.

In addition to the above, the responsibilities and duties of the company directors are explicitly stated in relevant company law and guidelines, similar to the Anglo-Saxon model [46]. These include: the board of directors (i) acts as the operational implementation organ of the company (ii) responsible for general shareholder’s meetings and can appoint and fire management and (iii) set up special committees such as auditing, remuneration and appraisal, strategy committees, etc.

3. Theoretical Framework

3.1. Corporate Sustainability Reporting (CSR)

Corporate sustainability reporting is a concept that promotes transparency and accountability to investors and other stakeholders regarding businesses’ contributions to global sustainable development goals [47]. CSR reports on an organization’s economic, environmental, and social responsibilities. Therefore, CSR is reporting the practices of an organization anchored on three main pillars: social, environmental, and economic sustainability. (i) Social sustainability reporting relates to disclosing information regarding labour practices and decent work, human rights issues, and product performance. (ii) Environmental sustainability reporting discloses information about energy, water, biodiversity, emissions, effluents and waste, compliance, and transport issues. (iii) Economic sustainability reporting includes information on revenues, taxes paid, operating costs, employee compensation, donations, and other investments. Therefore, corporate sustainability reporting allows investors and stakeholders to be acquainted with the sustainability strategies regarding the social, economic, and environmental structures of businesses toward sustainable goals [48].

3.2. Agency Theory

Agency theory is a principle that explains and resolves disputes between principals and agents. A conflict between a principal and an agent may arise due to a conflict of interest, opportunistic behaviour, and information asymmetry, creating a principal-agent problem [17]. The principle suggests reducing management's discretionary decision-making function to align this principal-agent goal. In other words, the board should monitor business executives to meet shareholder interests [49]. This theory, therefore, suggests that boards should be designed to advance the disclosure of a company's financial and non-financial information [13,17].

3.3. Resource Dependency Theory (RDT)

Resource dependency theory explains how a company's external resources affect its performance. Therefore, the theory posits that having the required competencies and skills makes the board a dynamic initiator of sustainability issues, enabling managers to acquire pro-social behaviour and enhance the firm's total value [30]. Therefore, the board's primary responsibility is to improve the company's economic, environmental, and social performance by judiciously allocating resources.

3.4. Legitimacy Theory

Legitimacy theory suggests that, for a company to continue to exist, it must act by meeting and adhering to society's expectations, norms, and rules [32]. Therefore, companies must repeatedly seek to ensure that they operate within the confines and norms of their respective societies, which implies that their operations are perceived as right and legitimate by outside parties. A company that goes contrary to this assertion may breach the social contract that the company has with society. This may lead to a revocation of the company's contract with the community.

3.5. Stakeholder Theory

According to Ortar (2018) [50], stakeholder theory forms the basis of sustainability reporting, which postulates that the success of a company depends on how it meets the needs of its stakeholders. The theory ensures that firms promote society's well-being, especially regarding its numerous environmental activities, such as pollution and noise. The theory further stipulates that firms should consider the interest of other stakeholders, such as employees, customers, governments, and suppliers [51]. This theory, therefore, requires that CEOs and the companies' boards act in unison to meet stakeholders' interests.

4. Literature and Hypotheses Development

4.1. Hypotheses Development

To show the relationship between board characteristics (such as board size, board independence, CEO duality, board meetings, and board committee) and CRS, we developed the following hypotheses as shown in Figure 1 below.

4.1.1. Board Size and CRS

The number of directors or members on a company's board is referred to as the board size. Although there is no universally accepted optimal board size, in China, the average size of a company's board is 9 members, with a range of 3 to 19 members. According to Mahmood et al. (2018) [15], a larger board size encourages sustainability reporting. These authors examined the top 100 Pakistani firms and concluded that a well-functioning board may increase profitability and thus increase reporting of its social and environmental impacts. Aksoy et al. (2020) [26] also found that a larger board size improves sustainability reporting in 63 Turkish firms. According to the authors, larger boards are more focused on the interests of shareholders and stakeholders in their decision-making process, which increases the likelihood of environmental, social, and governance (ESG) disclosure [13]. In

other words, larger boards are more sympathetic to stakeholders' concerns and demands and, thus, are more likely to disclose social and environmental impacts [52].

Other studies show the opposite. A study by Htay et al. (2012) [11], for example, found that smaller boards disclose more social and environmental information. According to these authors, smaller board sizes are more likely to make decisions faster, pay more attention to details, and disclose information much faster to stakeholders. Also, according to Lipton and Lorsch (1992) [53] and Jensen and Meckling (2019) [54], larger board sizes are disadvantageous because they are associated with weaker monitoring traits and a slower rate of decision-making, which impacts their disclosure of relevant information to stakeholders. This suggests that there are inconsistencies and conflicting findings in the existing literature regarding the role of board size on CSR. Furthermore, these studies concentrate on one or two pillars of CSR. However, given that several studies have established a positive relationship between larger board size and sustainability reporting, we hypothesize that:

Hypothesis 1 (H1): *Large board size may have a positive relationship with CSR.*

4.1.2. Board Independence and CSR

Independent directors are those whose decisions are not influenced by any other member of the company's board of directors. Thus, according to agency theory, independent directors on the board promote effective board supervision. Moreover, independent directors are more effective at advocating for strict adherence to the law and policies. Therefore, the company law of China requires one-third of the board of directors to be independent members.

Empirically, prior literature found that independent (or outside) board members are more likely than inside directors to advance economic, environmental, and social issues. Therefore, Jizi et al. (2014) [52] discovered that board independence and CSR disclosures are positively correlated in publicly-listed United States banks. In addition, Husted and de Sousa-Filho (2019) [13] found that board independence promotes sustainability reporting in 176 Latin American publicly-listed companies. Similarly, Wang, et al. (2016) [55], who examined 82 empirical studies to determine a link between board characteristics and CSR, found a link between board independence and sustainability reporting.

Similarly, Endrikat et al. (2021) and Wang et al. (2016) [23,55] examined 82 empirical studies to clarify the relationship between board characteristics and CSR and confirmed a positive relationship between them. However, contrary to popular belief, Mahmood et al. (2018) [15] believe that board independence does not improve corporate sustainability reporting. As a result, the relationship between board independence and corporate sustainability is not conclusive, and our paper examines this issue critically using a more comprehensive measure of CRS for Chinese publicly-listed firms. We hypothesized based on the literature that:

Hypothesis 2 (H2): *The presence of independent directors on corporate boards is positively related with CSR.*

4.1.3. CEO Duality and CSR

CEO duality is when a company's CEO serves as both its president and chairman of the board. This practice allows the CEO to exercise significant executive power in a more unified manner, which is beneficial for firms that need to make timely decisions about critical business transactions. Furthermore, because the same person holds the two most powerful positions in the company, the person may be able to use his influence to improve the company's performance while bridging the gap between management and the board through information disclosure. As a result, according to Jizi et al. (2014) [52], CEO duality promotes CSR.

However, some evidence in the literature points to the fact that a CEO who acts as both the company's president and its board chairman tends to be more focused on profit maximization and shows less attention to social and environmental issues. For instance, Husted and de Sousa-Filho (2019) [13] found a negative correlation between CEO duality and ESG disclosure when they studied 176 publicly listed companies in Latin America. Similarly, Shahbaz et al. (2020) [18], studied global energy firms and discovered that CEO duality is negatively related to ESG disclosure. According to Shahbaz et al. (2020) [18], if the same person occupies both positions, they become more powerful and can override the board, making the board's decision-making process and the firm's operations challenging. There will also be executive compensation conflicts and abuse of corporate governance structures, minimizing the credibility and effectiveness of the audit committee's function. Therefore, we anticipate a negative relationship between CEO duality and sustainability disclosure in this study, and therefore, we hypothesize:

Hypothesis 3 (H3): *CEO duality is negatively associated with CSR.*

4.1.4. Board Meetings and CSR

Every now and then, the board of directors will convene for a formal meeting to discuss and make important decisions, such as hiring new executives, firing underperforming ones, and deciding how to best report to shareholders and other stakeholders on the company's financials, environmental impacts, and social welfare. According to Jizi et al. (2014) [52] and Jizi (2017) [19], the frequency of their meetings demonstrates the board's stability and promotes greater transparency in information disclosure to stakeholders. Therefore, these studies concluded that board meeting frequency influences CSR, suggesting that more frequent board meetings may increase CSR. Therefore, the study hypothesizes that:

Hypothesis 4 (H4): *There is a positive relationship between a board meeting and CSR.*

4.1.5. Board Committees and CSR

Sustainability should not be treated as a separate activity, but rather be incorporated into the company's overall approach to defining its mission, crafting its strategy, and addressing the threats and opportunities. Sustainable development should become ingrained in the company's operational practices, as well as its decision-making and culture. To build a sustainable world as well as ensure the long-term success of businesses, sustainability must be done correctly. For this reason, it is necessary to establish a board sustainability committee charged with monitoring environmental and social sustainability.

The board's sustainability committee assesses major policies and performance and provides guidance on sustainability-related issues. A board-level sustainability committee will advocate for strategies and implementation plans related to economic, environmental, and social issues to effectively execute sustainability performance and information reporting. According to Chams and García-Blandón (2019) [56], the board committees have a significant impact on sustainability performance and reporting. The board sustainability committee is therefore important for CSR, and we propose that (Figure 1):

Hypothesis 5 (H5): *There is a positive relationship between board committees and CSR.*

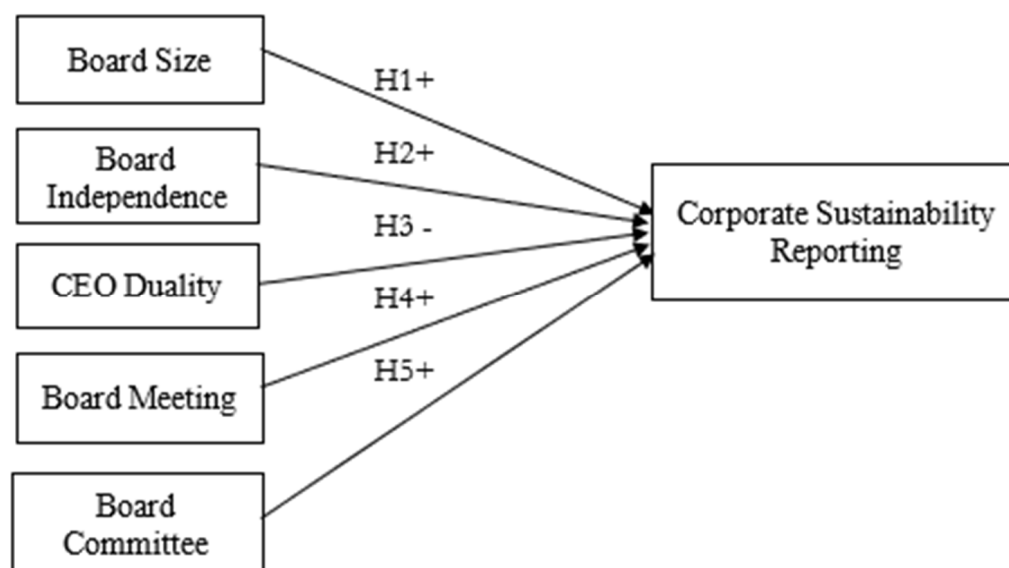


Figure 1. Research framework and hypothesized relationships.

5. Research Methodology

5.1. Data Collection and Selection Criteria

The study uses annual reports, financial statements, and reports on corporate sustainability from a sample of companies. The initial sample for this research was about 4600 companies listed on the Shanghai and Shenzhen stock exchanges. However, due to missing data for some companies, about 3600 of them were eliminated. Since extreme values may distort statistical results, over 250 potential outliers were removed from both tails, leaving behind a final sample of 703 companies. Therefore, the final sample used for analysis is made up of 9842 firm-year observations from 703 companies over 14 fiscal years, from 1 January 2006 to 31 December 2019. Data on sustainability performance is manually collected from annual and standalone corporate sustainability reports. In contrast, financial data are extracted from audited financial statements on the China Stock Market and Accounting Research (CSMAR) database.

5.2. Measurements of Variables

5.2.1. Dependent Variable

Following studies like Orazalin & Mahmood (2018) [37], the current study employs a dichotomous approach to measure the various components of corporate sustainability reporting. When an item is disclosed, it is given a value of one; otherwise, zero. The Global Reporting Initiative (GRI 3.1) gives countries scores for how well they report on their economic, environmental, and social sustainability. The GRI 3.1 offers 82 specific performance indicators under three disclosure categories, i.e., 9 economic indicators, 28 environmental indicators, and 45 social indicators (see details in Table 1). For each indicator, if an item is disclosed, it receives a value of one; otherwise, zero. By summing the totals of each item, the scores for sustainability reporting are determined. The score of each firm is calculated as the sum of disclosed items divided by the number of items in each category: economic, environmental, and social. The composite sustainability reporting index is calculated as the sum of the total number of items in economic, environmental, and social sustainability reports divided by eighty-two items. The final results are expressed as a percentage ranging from 0 to 100.

Table 1. Summary of indicators within the three-sustainability reporting dimension.

S #	Dimension	Category	Aspect	Indicators
1	Economic	Economic performance indicators	Economic performance, market presence, indirect economic impacts, and procurement practices.	9
2	Environmental	Environmental performance indicators	Materials, energy, water, biodiversity, emissions, effluents and waste, products and services, compliance, transport, overall, supplier environmental assessment, and environmental grievance mechanisms.	28
3	Social	Labour practice and decent work performance indicators	Employment Labour/Management relations Occupational health and safety training and education Diversity and equal opportunity Equal remuneration for women and men.	15
		Human rights performance indicators	Investment and procurement practice Non-discrimination Freedom of association and collective bargaining Child labour, forced and compulsory labour, security practices, Indigenous rights, assessment, remediation	11
		Social performance indicators	Local communities, corruption, public policy, anti-competitive behaviour, compliance	10
		Product responsibility performance indicators	Customer health and safety Product and service labelling Marketing and communications Customer privacy Compliance	9
Total				82

Source: Global Reporting Initiative (GRI 3.1).

5.2.2. Independent variables

Our independent variables include board characteristics like board size, board membership, CEO duality, board meetings, and the number of committees on the board. The board size is measured as the total number of directors on the firm's board. The CEO duality is measured as a binary variable: "1" when the CEO also serves as chairman and "0" otherwise. The variable for independent board members is measured as the proportion of non-executive board members on the board. The number of board meetings is measured as the total number of regular or emergency meetings held in the year. The board committee is measured as the total number of committees on the board.

Beyond the explanatory variables, we introduce some theoretically relevant control variables relating to CSR, and these include firm size, leverage, profitability (i.e., return on asset), firm age, and the BIG 4 (i.e., whether any of the big four audit companies audit their accounts). See details for all variables in Table 2.

Table 2. Summary of variables measurements.

Category	Sub-Category	Abrev	Description/Measurement
Dependent Variable	Corporate sustainability reporting index	CSRI	The metric assigns “1” to each item in the sustainability report if the item is disclosed, and “0” for a non-disclosed item.
	Economic sustainability reporting	ECON	All 9 items listed in the economic indicators are assigned “1” if the item is disclosed and “0” if it is non-disclosed.
	Environmental sustainability reporting	ENV	All 28 items listed in the environmental indicators are assigned “1” if the item is disclosed and “0” if it is non-disclosed.
	Social sustainability reporting	SOC	All 45 items listed in the environmental indicators are assigned “1” if the item is disclosed and “0” if it is non-disclosed.
Independent Variables	Board size	BS	Total number of directors on the board
	Board independence	BIND	The proportion of independent directors on the board
	CEO duality	CEO_Dual	Used a binary variable where “1” indicates whether the CEO serves as chairman, and “0” indicates otherwise.
	Board meeting	BOD_Mtg	Number of times meetings are held
	Board’s committees	BC	Number of committees on the board
	Return on assets	ROA	Net profit/total assets
Control Variables	Leverage	LEV.	Total liabilities are divided by the total assets
	Companies age	Fage	It is calculated as the number of years elapsed since its establishment
	Company size	Size	The log of the total assets at the end of the period
	BIG 4	BIG4	Reports of companies audited by any BIG 4 auditing companies are denoted as “1” and otherwise “0”

5.3. Model Specification

The general model for this study follows the existing literature, specifically the work of Orazalin and Mahmood (2018) [37], where sustainability reporting practices are modelled as a function of possible underlying factors. In our case, we set CSR as a function of board characteristics with some controls.

$$CSRI_{it} = \alpha_i + \beta_1 BSIZE_{it} + \beta_2 BIND_{it} + \beta_3 CDUAL_{it} + \beta_4 BM_{it} + \beta_5 BC_{it} + \beta_6 CSIZE_{it} + \beta_7 LEV_{it} + \beta_8 ROA_{it} + \beta_9 BIG4_{it} + \varepsilon_{it} \quad (1)$$

where, $\beta_1, \dots, \beta_9$ are the coefficients of board size (*BSIZE*), board independence (*BIND*), CEO duality (*CDUAL*), board meeting (*B.M.*), board committee (*B.C.*), company size (*CSIZE*), leverage (*LEV*), return on asset (*ROA*), and *BIG4* auditing firms (*BIG4*). *CSRI* is the composite corporate sustainability reporting index, ε represents the error term, *i* is the cross-section, and *t* denotes the time series.

With the above model, we investigate the effect of board characteristics on the three pillars of corporate sustainability reporting—economic, environmental, and social. Thus, we have the following models:

$$ECON_{it} = \alpha_i + \beta_1 BSIZE_{it} + \beta_2 BIND_{it} + \beta_3 CDUAL_{it} + \beta_4 BM_{it} + \beta_5 BC_{it} + \beta_6 CSIZE_{it} + \beta_7 LEV_{it} + \beta_8 ROA_{it} + \beta_9 BIG4_{it} + \varepsilon_{it} \quad (2)$$

$$ENV_{it} = \alpha_i + \beta_1 BSIZE_{it} + \beta_2 BIND_{it} + \beta_3 CDUAL_{it} + \beta_4 BM_{it} + \beta_5 BC_{it} + \beta_6 CSIZE_{it} + \beta_7 LEV_{it} + \beta_8 ROA_{it} + \beta_9 BIG4_{it} + \varepsilon_{it} \quad (3)$$

$$SOC_{it} = \alpha_i + \beta_1 BSIZE_{it} + \beta_2 BIND_{it} + \beta_3 CDUAL_{it} + \beta_4 BM_{it} + \beta_5 BC_{it} + \beta_6 CSIZE_{it} + \beta_7 LEV_{it} + \beta_8 ROA_{it} + \beta_9 BIG4_{it} + \varepsilon_{it} \quad (4)$$

where, *ECON* represents the economic corporate sustainability reporting index, *ENV* denotes the environmental corporate sustainability reporting index, and *SOC* means the social corporate sustainability reporting index.

5.4. Econometric Modelling

5.4.1. Stationarity Test

The methodological framework of this study starts with an estimate of the unit root to figure out the order of variable integration. Therefore, by adopting the unit root tests developed by Im, Pesaran, and Shin (2003) [57] and Levin, Lin, and Chu (2002) [58], we examine the unit root properties of each series variable. These two unit root tests were chosen because they provide accurate estimates and are consistent with most previous research. The following model is used to conduct these tests:

$$\Delta Y_{it} = \varnothing_i + \alpha_i \chi_{i,t-1} + \delta_i t + \sum_{j=1}^k \gamma_{ij} \Delta \chi_{i,t-j} + \mu_{it} \quad (5)$$

where, Y_{it} is the dependent variable; $\varnothing_i$, α_i , δ_i , and γ_{ij} are model parameters; Δ is the first difference operator; μ_{it} is the error term; i and t are the country and period in question, respectively. The Im, Pesaran, and Shin (2003) [57] (IPS) estimates the null hypothesis $H_0 : \alpha_1 = 0$ for some i , against the alternative that all series are stationary, $H_1 : \alpha_1 < 0$ for some i . This hypothesis' rejection adds to the evidence that the series is stationary.

5.4.2. Cross-Section Dependence

Second, following Pesaran (2007) [59], we examine the cross-sectional dependence of our variables using the following equation:

$$\Delta Y_{it} = \pi_i Y_{i,t-1} + \gamma_i Z_{it} + \sum_{j=1}^{k-1} \theta_{ij} Y_{i,t-j} + \varepsilon_{it} \quad (6)$$

Z_{it} is a deterministic component in this scenario, and $\sum_{j=1}^{k-1} \theta_{ij} Y_{i,t-j}$ is treated as an ADF test. As a result, when objects i share the same factors, it can be considered cross-sectional. By defining that, we can obtain the following:

$$\varepsilon_{it} = \theta_i f_i + u_{it} \quad (7)$$

θ_i denotes that each individual has a distinct impact, and u_{it} denotes no cross-section or autocorrelation in the data. Equation (7) substitutes into Equation (6) to get the following results:

$$\Delta Y_{it} = \pi_i Y_{i,t-1} + \gamma_i Z_{it} + \sum_{j=1}^{k-1} \theta_{ij} Y_{i,t-j} + \theta_i f_i + u_{it} \quad (8)$$

With Equation (8), we test whether or not there is cross-sectional dependence among variables based on the hypothesis: $H_0: \theta_i \neq 0$ and $H_1.A.: \theta_i = 0$.

5.4.3. Cointegration Test

After determining the order in which the variables should be integrated, the next step is to investigate their long-term relationship. Therefore, following Kao (1999) [60], we test for cointegration in our variables using the following equation:

$$y_{it} = X'_{it}\beta_i + z'_{it}\gamma_i + \varepsilon_{it} \quad (9)$$

The test focused on the covariates in X'_{it} that are not cointegrated. The cointegrating phenomenon vector, denoted by the β_i , can vary between different panels. The γ_i vector contains the Z'_{it} coefficients, known as deterministic terms, to control panel-specific effects and linear time trends. ε_{it} is the error term.

Kao (1999) identifies the assumptions of the cointegrating vector from Equation (6) with $\beta_i = \beta$; consequently, the slope coefficients of the panels are standard. Thus, Kao (1999) [60] proposes five types of tests from the Dickey-Fuller regression, including the Modified Dickey-Fuller t , Dickey-Fuller t , Augmented Dickey-Fuller t , Unadjusted modified Dickey-Fuller t , and Unadjusted Dickey-Fuller t :

$$\hat{e}_{it} = \rho \hat{e}_{i, t-1} + v_{it} \quad (10)$$

where ρ is the estimated residuals' common auto regression parameter. Meanwhile, here is the Augmented Dickey-Fuller regression:

$$\hat{e}_{it} = \rho \hat{e}_{i, t-1} + \sum_{j=1}^k \rho_j \Delta \hat{e}_{i, t-j} + v_{it} \quad (11)$$

Interestingly, in this method, ρ reveals the number of lagged difference terms.

5.4.4. Estimation Strategy

After the long-run relationship between the series is confirmed, the long-run model can be estimated using the ordinary least squares (OLS) technique. However, the OLS has been linked to second-order asymptotic bias and serial correlation [61]. Stock and Watson (1993) [62] developed the dynamic ordinary least squares (DOLS) while Phillips and Moon (1994) [63] proposed the fully modified ordinary least squares (FMOLS) to address these issues. Therefore, both the panel DOLS and the FMOLS can be used to deal with the small sample bias in OLS estimation. In this paper, we use both methods because of their estimation power for correcting simultaneity bias and serial correlation challenges. The primary distinction between the two approaches is how autocorrelation in regression is corrected. While FMOLS can handle Newey-West correction, DOLS can handle more lag and lead variables. Pedroni (2001) [61] developed the method for estimating coefficients, which is now the standard method for calculating the effects of long-run variables and is as follows:

$$\hat{\beta}_{FMOLS} = \left(\sum_{i=1}^N \hat{L}_{22i}^{-1} \sum_{t=1}^T (X_{it} - \bar{X}_i)^2 \right)^{-1} \sum_{i=1}^N \hat{L}_{11i}^{-1} \hat{L}_{22i}^{-1} \left(\sum_{t=1}^T (X_{it} - \bar{X}_i) y_{it}^* - T \delta_i \right) \quad (12)$$

in which

$$y_{it}^* = (y_{it} - \bar{y}_i) - \left(\frac{\hat{L}_{21i}}{\hat{L}_{22i}} \right) \Delta X_{it} + \left(\frac{\hat{L}_{21i} - \hat{L}_{22i}}{\hat{L}_{22i}} \right) \beta (X_{it} - \bar{X}_i) \quad (13)$$

and we denote δ_i as

$$\delta_i \equiv \hat{L}_{21i} + \hat{\Psi}_{21i}^0 - \left(\frac{\hat{L}_{21i}}{\hat{L}_{22i}} \right) (\hat{L}_{21i} + \hat{\Psi}_{21i}^0) \quad (14)$$

In order to clarify this, we refer to Ψ as an asymptotic covariance matrix for long-run variance and as dynamic covariance. In addition, L is a lower triangular matrix with partition calculation. Then, the DOLS estimator is employed, which takes the following form:

For clarity, Ψ will be referred to as a dynamic covariance matrix and an asymptotic covariance matrix for long-run variance. Furthermore, L is a lower triangular matrix that includes partition calculations. The DOLS estimator, which looks like this, is then used.

$$y_{it} = \beta_i X'_{it} + \sum_{j=-k}^k \theta_{ij} \Delta X_{i,t+k} + \gamma_{1i}' D_{1i} v_{it} \quad (15)$$

K represents the total number of leads or lags used in the models. Figure 2 below highlights the research methodology adopted in this study.

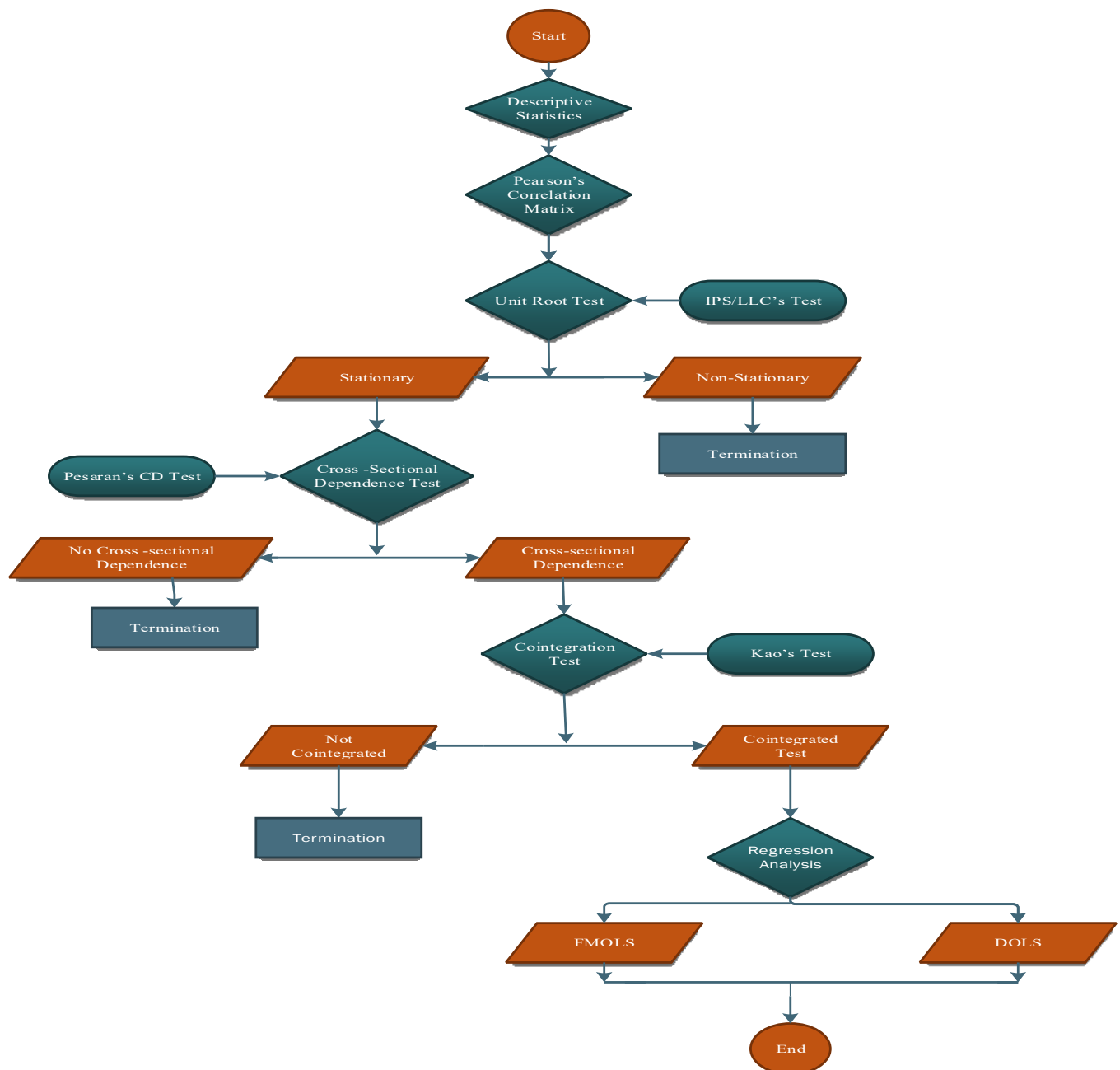


Figure 2. Methodological framework of the study.

6. Results and Discussion

6.1. Preliminary Tests

Table 3 presents the summary statistics. From Table 3, the composite CSR has a mean of 63.04 and a standard deviation of 48.27, suggesting that more than half of Chinese firms disclose sustainability reports. A mean value of nine for board size suggests that, on average, a board has nine members. A mean value of 9 for the number of board meetings per year indicates that firms typically meet nine times on average annually. For CEO duality, our statistics show that 13.9% of companies have the same person occupying the CEO and board's chairmanship positions. The table also shows that, on average, these firms have been in existence for about 19 years.

Table 3. Descriptive statistics.

Variable	Obs	Mean	Std. Dev.	Min	Max	VIF
CSR	9842	63.035	48.273	0.215	79.132	
Economic	9842	24.304	42.894	0.275	76.437	
Environmental	9842	59.114	49.165	0.043	92.063	
Social	9842	60.770	48.829	0.422	92.480	
Board Size	9842	9.343	2.021	3	20	3.19
Bind	9842	3.387	0.749	1	8	3.21
CEO Duality	9842	0.139	0.346	0	1	1.03
Board Mtg	9842	9.413	4.223	2	56	1.06
Board Cttee	9842	3.856	0.729	0	8	1.07
Company Size	9842	9.589	0.637	7.26	13.45	1.62
Lev	9842	9.671	210.176	0	13,252.58	1.00
ROA	9842	0.107	2.293	−19.55	187.47	1.01
Company Age	9842	19.486	5.301	2	41	1.33
Big4	9842	0.088	0.283	0	1	1.18

In the next stage, we carried out some preliminary tests. First, we used Pearson's product-moment correlation coefficient technique to assess the relationship between the explanatory variables. Table 4 summarizes the correlation coefficient. The results indicate a mixture of small and moderate correlations among the variables. All the variables showed a low correlation with corporate sustainability reporting and its dimensions, except for company size, which recorded a moderate correlation with total corporate sustainability reporting and the other pillars. At the same time, firm age also recorded a moderate correlation with CSR and the other two pillars (environmental and social). Although there was a weak and moderate correlation among the variables, most of the predictor variables, the board size, board independence, board meeting, and board committee, are statistically significant and positively correlated with corporate sustainability reporting and its pillars, except for CEO duality, which reported statistical insignificance and a negative correlation with total corporate sustainability reporting and two pillars (environmental and social). However, it recorded a statistically significant negative correlation with economic reporting.

The correlation matrix confirms that there is no issue of multicollinearity, since the values for all variables are lower than 0.90 [64]. Moreover, the values for VIF, as shown in Table 3, are comparatively small and below 5. This justifies the fact that multicollinearity among the study variables was non-existent.

Next, we used Im, Pesaran, and Shin 2003 [57] (IPS) and Levin, Lin & Chut 2002 [58] (LLC) to test for a unit root. Table 5 illustrates variables at either significance or insignificance at level. Thus, we tested at first deference. At (1), all the variables are stationary, and therefore, we reject the null hypothesis of non-stationarity.

Table 4. Pearson's Correlation coefficient.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
(1) CSR	1.000													
(2) Economic	0.434 *	1.000												
(3) Environmental	0.721 *	0.436 *	1.000											
(4) Social	0.753 *	0.455 *	0.872 *	1.000										
(5) Board Size	0.028 *	0.150 *	0.038 *	0.041 *	1.000									
(6) Bind	0.064 *	0.179 *	0.075 *	0.076 *	0.822 *	1.000								
(7) CEO Duality	−0.009	−0.037 *	−0.009	−0.005	−0.155 *	−0.115 *	1.000							
(8) Board Mtg	0.096 *	0.081 *	0.081 *	0.098 *	0.003	0.041 *	0.006	1.000						
(9) Board Cttee	0.142 *	0.094 *	0.136 *	0.141 *	0.092 *	0.138 *	0.002	0.085 *	1.000					
(10) Firm Size	0.323 *	0.376 *	0.338 *	0.328 *	0.335 *	0.363 *	−0.084 *	0.214 *	0.136 *	1.000				
(11) Lev	−0.009	−0.017	−0.014	−0.007	−0.010	−0.013	0.019	0.002	−0.011	−0.038 *	1.000			
(12) ROA	0.024 *	0.012	0.027 *	0.025 *	0.015	0.023 *	−0.009	0.019	−0.003	−0.040 *	−0.001	1.000		
(13) Firm Age	0.478 *	0.171 *	0.450 *	0.448 *	−0.076 *	−0.019	0.036 *	0.084 *	0.182 *	0.180 *	0.003	0.028 *	1.000	
(14) Big4	0.101 *	0.172 *	0.112 *	0.103 *	0.161 *	0.162 *	−0.041 *	0.062 *	−0.003	0.377 *	−0.003	0.044 *	0.029 *	1.000

Note: * Significance at 10% level.

Table 5. Results for the unit root test.

TEST	IPS		LLC	
	Level	1st Difference	Level	1st Difference
CSR	0.09534	−14.6182 ***	−8.053 ***	−29.956 ***
Economical	−7.45071 ***	−19.6617 ***	−20.219 ***	−16.220 ***
Environmental	−1.56255	−15.6631 ***	−10.951 ***	−26.581 ***
Social	0.99421	−12.0233 ***	−6.915 ***	−26.821 ***
Board Size	−10.8785 ***	−19.9493 ***	−53.763 ***	−29.002 ***
BIND	−2.36781 ***	−10.2294 ***	−16.434 ***	−19.804 ***
CEO Dual	−1.60287	−8.31628 ***	−10.834 ***	−16.523 ***
Board Mtg	−13.8279 ***	−35.8510 ***	−29.021 ***	−49.589 ***
Board Cttee	−6.88683 ***	−17.6486 ***	−16.113 ***	−13.934 ***
ROA	10.8823	−7.47590 ***	6.806	−72.263 ***
Firm Size	31.7197	12.8127 ***	40.799	−35.575 ***
Firm Age	0.65544	−0.27031 ***	64.471	−1.093 ***
LEV.	−13.6533 ***	−31.9785 ***	18.086	−98.557 ***
BIG4	1.34573	−2.78607 ***	−3.676 ***	−8.366 ***

Note: *** Significance at 1% level.

In the next step, we carried out a cross-sectional dependence test using the CD test proposed by [65]. Table 6 shows the results for the CD test and the probability values for the tests are significant at the 1% level. Therefore, the null hypothesis of no cross-sectional dependence is rejected.

Based on the assumption of cross-section dependence, we went ahead with the cointegration test. As shown in Table 7, the probability results for models 1–4 are significant at a 1% level. Therefore, the null hypothesis of no cointegration is rejected.

Table 6. Results of cross-sectional dependence.

Variable	CD-Test	p-Value	Corr	Abs (Corr)
CSR	946.912	0.000	0.51	0.52
Economic	98.532	0.000	0.05	0.10
Environmental	855.582	0.000	0.46	0.47
Social	864.17	0.000	0.46	0.48
Board Size	45.128	0.000	0.02	0.23
BIND	6.416	0.000	0.00	0.09
CEO Dual	2.31	0.021	0.00	0.05
Board Mtg	94.738	0.000	0.05	0.25
Board Cttee	100.084	0.000	0.05	0.09
Firm Size	654.22	0.000	0.35	0.53
LEV	8.559	0.000	0.00	0.21
ROA	174.588	0.000	0.09	0.34
Firm Age	1837.193	0.000	0.99	0.99
BIG4	0.607	0.005	0.00	0.00

Notes: The test statistical assumes a normal standard distribution N (0, 1).

Table 7. Results of the cointegration test.

Test	CSR		Economic CSR		Environmental CSR		Social CSR	
	t-Statistic	Prob.	t-Statistic	Prob.	t-Statistic	Prob.	t-Statistic	Prob.
ADF	−20.1442	0.0000	−25.6959	0.0000	−19.926	0.0000	−18.3040	0.0000

6.2. Hypothesis Testing

Having conducted the above preliminary tests, we examined the relationship between board characteristics and corporate sustainability reporting. The analysis is set up as follows: (i) The first section discusses the outcome of the relationships between board characteristics and the composite CRS; and (ii) The second section focuses on the three pillars of corporate sustainability reporting (i.e., economic, environmental, and social) and their connection to board characteristics. To address the issue of endogeneity, throughout the analysis, we employed both the FMOLS and DOLS.

6.2.1. Section 1: The Effect of Board Characteristics and Composite CSR

Table 8 presents the results for the effects of board characteristics on the composite CSR. To thoroughly examine these effects and ensure the consistency of the outcome of our key variables on CSR, we started by estimating these variables individually in the models. Therefore, beginning with board size while controlling for factors such as company size, age, leverage, ROA, and the BIG4, the results in column (I) of Table 8 show a positive and significant relationship between board size and the composite CSR, where (β_1 0.052 and 0.00014) in both FMOLS and DOLS, respectively. In other words, companies with larger board sizes are more likely to disclose information on sustainability performances and impacts to stakeholders. On that basis, hypothesis 1 is accepted, confirming the study's earlier prediction. Therefore, larger boards increased companies' sustainability information disclosure. It is expected that larger boards will engage in broader decision-making perspectives, requiring greater debate and negotiations. As observed in other markets, similarly, larger boards in Chinese companies represent different stakeholder groups and concerns, making them more likely to report more about their CSR performance to satisfy these stakeholder needs. Thus, it is perceived that larger boards seemed more geared towards stakeholders' interests and consequently viewed corporate sustainability performances and impacts as part of the general corporate strategy. These findings refute the idea of having a smaller board

size. However, these results are consistent with the findings cited from studies conducted in other regions, especially in Latin America, Europe, Asia, and other emerging markets. For instance, Mohamood et al. (2018) [15] observed that larger board size helps to improve sustainable information disclosure. Similarly, Husted and de Sousa-Filho (2019) [13] confirm that larger boards effectively increase ESG disclosure. However, it contradicts those in Htay et al. (2012) [11], who concluded that a limited size of the board performs better when it comes to social and environmental information disclosure.

In column II, we consider the second independent variable, board independence (BIND), which positively correlates with the composite CSR, (β_2 0.0138 and 0.0060) in both FMOLS and DOLS. According to these results, the board's level of independence plays a principal role in the extent of information disclosure about sustainability initiatives with stakeholders. Therefore, a corporate board with a higher proportion of non-executive directors is more likely to be associated with higher sustainability information reporting, since outside directors are more stakeholder-driven than inside directors. These findings are equally consistent with other studies. For example, Jizi et al. (2014) [52] found a positive connection between independent board directors and CSR disclosure. Other related studies also demonstrate a significant positive relationship between board independence and social sustainability reporting [23,55]. Additionally, studies such as Husted and de Sousa-Filho (2019) [13] also confirm that board independence helps increase ESG disclosures. Surprisingly, our study contradicts the study by Mahmood et al. (2018) [15], who concluded that board independence is negatively correlated with sustainable information disclosure, and Liu et al. (2020) [24], who also suggests that independent directors impede sustainable operations.

However, in column III, CEO duality negatively correlates with composite CSR. According to the results, where (β_3 − 0.0328 and − 0.0181), it is implied that an increase in CEO duality reduces CSR. CEO duality is when the CEO, besides running the organization at the highest level, also holds the position of chairman of the board. In this situation, the CEO has a lot of executive power and can use it to make better business decisions more quickly. However, the demerit is that a CEO serving as the board chairman may create a conflict of interest, abuse power, and probably minimize the effectiveness of what the committee does and reports as a sustainability report. This study's findings confirm the result of Husted & de Sousa-Filho (2019) [13], who show that CEO duality is negatively related to ESG disclosures. However, our findings are also inconsistent with Jizi et al. (2014) [52], who point out that CEO duality positively impacts CSR disclosures.

Board meetings promote CSR disclosures in column IV, where (β_4 0.0009 and 0.0003). The results suggest that the board's ability to address stakeholders' CSR concerns depends on how frequently they meet to make those critical decisions. Thus, the frequency at which board meetings are held increases the likelihood of reporting sustainable information to stakeholders. This finding confirms similar studies conducted in the U.S., where Jizi (2017) [19] and Jizi et al. (2014) [52] observed that board meetings improve CSR.

We also observed a positive relationship between board committees and CSR in column V with (β_5 0.01658 and 0.01322). This implies that board committees promote CSR. Possible explanations for such a result include the critical role of the presence of sustainability committees on the board to help scrutinize issues and address policies and standards that may have formed a primary component, increasing the likelihood of reporting sustainable information to stakeholders. Similar results are found in the study conducted by Chams and García-Blandón (2019) [56] in the U.S.

Finally, in column VI, we considered all the explanatory variables (i.e., board characteristics) in one model to improve the model identification regarding parameter estimates and interpretation of what types of board characteristics improve CSR. From the table, the results for each variable, whether estimated separately in columns I–V, remain consistent with those jointly estimated in column VI. Similarly, the results for control variables remain consistent throughout the models, i.e., from columns I–VI. The results for the controls show

that company size, ROA, and firm age positively relate with CSR, while leverage and BIG4 negatively impact CSR.

6.2.2. Section II: The Effect of Board Characteristics and the Three Components of CSR Performance

Unlike the previous section, where the investigation focused on the effect of board characteristics on aggregate CRS, in this section, we examine the effect of board characteristics on the three pillars of CSR. Therefore, Table 9 shows the results for the economic CSR—board characteristics nexus. Table 10 presents results for the relationship between environmental CSR and board characteristics, while Table 11 discloses results on social CSR and board characteristics.

As in Tables 8–11 examine the independent variables (i.e., the board characteristics) separately in each model, starting with board size. In each model, we also control for factors like firm size, age, leverage, ROA, and the BIG4. In Tables 9–11, the results show that board size (β_1 0.0116 and 0.018, 0.0053 and 0.00021, 0.0032 and 0.00355) has a positive and significant relationship with the three pillars of CSR: economic, environmental, and social, respectively. That is, an increase in board size increases economic, environmental, and social CSR. Results for economic CSR suggest firms with larger boards are more motivated to disclose their economic performance, i.e., information on the distribution of revenue, employee compensation, taxes, and infrastructure investments. These results are consistent with Mahmood et al. (2018) [15], who observed a similar effect. The outcome for environmental CSR indicates that firms with a larger board size are more likely to report on environmental sustainability issues, such as waste and pollution management, to stakeholders. This is related to the fact that corporations are expanding their boards to include members with knowledge and experience in sustainable development for a variety of reasons. Customers and affected communities are becoming more aware not only of environmental sustainability concerns and implications, but also of their power to effect change, which is why investors are increasingly demanding information about these activities. Furthermore, compliance requirements for environmental sustainability issues are increasing, and many international regulators have stringent environmental sustainability disclosure rules. Additionally, our findings support Husted and de Sousa-Filho's (2019) [13] work in Latin America, which confirms that a larger board size effectively increases ESG disclosure. Similar results were found by Masud et al. (2018) [16] in their study of 88 listed companies in South Asia, who concluded that board size enhances environmental disclosure. However, it contradicts the study by Aliyu (2019) [22], who observed that a larger board size is insignificantly related to corporate environmental reporting in Nigerian non-public listed companies.

For social CSR, this result indicates that firms with larger boards encourage disclosure of social issues, thereby ensuring healthy labour practices, human rights protection, and product responsibility. These results are consistent with those of Mahmood et al. (2018) [15], who studied 100 firms in Pakistan and observed that a larger board size helps improve sustainable information disclosure. Liu and Zhang (2017) [5] researched social responsibility information disclosure in heavy-pollution industries and concluded that a larger board size influences CSR disclosure in China. This assertion supports the findings of this study. In the case of social sustainability reporting, Endrikat et al. (2021) [23], who sampled 82 empirical studies to clarify a relationship between board characteristics and CSR, concluded that board size influences CSR directly. However, it goes against what Htay et al. (2012) [11] found, which was that a smaller board size does better when it comes to sharing social and environmental information.

Table 8. FMOLS and DOLS regression results for the relationship board characteristics and composite CSR.

Variables	Column (I)		Column (II)		Column (III)		Column (IV)		Column (V)		Column (VI)	
	FMOLS	DOLS	FMOLS	DOLS	FMOLS	DOLS	FMOLS	DOLS	FMOLS	DOLS	FMOLS	DOLS
Board Size	0.0052 ***	0.00014 ***									0.0030 ***	0.0031 ***
BIND			0.0138 ***	0.0060 ***							0.0078 ***	0.0131 ***
CEO duality					−0.0328 ***	−0.0181 ***					−0.0293 ***	−0.018 ***
Board Mtg							0.0009 **	0.0003 **			0.00153 ***	0.00027 ***
Board C'ttee									0.01658 ***	0.01322 ***	0.01882 ***	0.0136 ***
Firm Size	0.1968 **	0.1402 **	0.1971 **	0.1411 **	0.19174 ***	0.1405 ***	0.1914 **	0.1398 ***	0.1865 ***	0.1387 ***	0.1981 ***	0.1392 ***
Leverage	-1.51×10^{-6}	-7.82×10^{-6}	-1.67×10^{-6}	-7.91×10^{-6}	-1.63×10^{-7}	-7.73×10^{-6}	-1.56×10^{-6}	-7.87×10^{-6}	-2.33×10^{-6}	-7.57×10^{-6}	-1.31×10^{-6}	-7.69×10^{-6}
ROA	0.0057 **	0.0008 ***	0.0057 **	0.00087 ***	0.0055 **	0.00083 ***	0.00553 **	0.00084 ***	0.0055 **	0.0007 ***	0.0059 ***	0.0008 ***
Firm Age	0.0257 ***	0.605 ***	0.0258 **	0.0681 ***	0.0258 **	0.0682 ***	0.0260 **	0.681 ***	0.02564 **	0.67763 ***	0.0253 ***	0.0678 ***
BIG4	−0.0111	−0.8791	−0.011980	−0.0515	−0.0113	−0.0523	−0.0117	−0.0515	−0.00593	−0.0525	−0.0063	−0.0532
Obs	9842	9842	9842	9842	9842	9842	9842	9842	9842	9842	9842	9842
Constant	−1.6916		−1.6965		−1.6978		−1.6831		−1.7032		−1.7545	
R Squared	0.11441	0.5887	0.11492	0.5887	0.1131	0.5888	0.11363	0.5887	0.1171	0.5889	0.11778	0.589

Notes: FMOLS denotes Fully Modified Least Squares, DOLS denotes Dynamic Ordinary Least Squares, BIND represents Board independence, ROA means Return on Assets, and the BIG4 is big four audit companies. ** Significance at 5% level. *** Significance at 1% level.

Table 9. FMOLS and DOLS regression results for the relationship between board characteristics and economic CSR performance.

Variables	Column (I)		Column (II)		Column (III)		Column (IV)		Column (V)		Column (VI)	
	FMOLS	DOLS	FMOLS	DOLS	FMOLS	DOLS	FMOLS	DOLS	FMOLS	DOLS	FMOLS	DOLS
Board Size	0.0116 ***	0.0018 ***									0.00203 ***	0.0061 ***
BIND			0.0402 ***	0.0182 ***							0.0417 ***	0.0268 ***
CEO Duality					−0.0103 ***	−0.0239 ***					−0.0034 ***	−0.0236 ***
Board Mtg							0.00207 ***	0.001 ***			0.001962 ***	0.0010 ***
Board C'ttee									0.0265 ***	0.0189 ***	0.0219 ***	0.0176 ***
Firm Size	0.2434 ***	0.1015 ***	0.2380 ***	0.0997 ***	0.2571 ***	0.1023 ***	0.261 ***	0.100 ***	0.2548 ***	0.099 ***	0.23976 ***	0.0971 ***
Leverage	−0.00002	-7.06×10^{-6}	−0.00002	-6.79×10^{-6}	−0.00002	-6.94×10^{-6}	−0.00002	-7.21×10^{-6}	−0.00002	-6.71×10^{-6}	−0.00002	-6.37×10^{-6}
ROA	0.00297 **	6.83×10^{-5} **	0.0027 ***	0.00015 **	0.0033 **	9.14×10^{-5} **	0.00337 **	7.75×10^{-5} **	0.00335 **	0.00015 **	0.0028 **	0.0002 **
Firm Age	0.00896 ***	0.0146 ***	0.0088 ***	0.0148 ***	0.0085 ***	0.0146 ***	0.0086 ***	0.0145 ***	0.0081 ***	0.0398 ***	0.00848 ***	0.0139 ***
BIG4	−0.0205	−0.0657	−0.0217	−0.0656	−0.0224	−0.0647	−0.0217	−0.0658	−0.0256	−0.0643	−0.02405	−0.0635
Observations	9842	9842	9842	9842	9842	9842	9842	9842	9842	9842	9842	9842
Constant	−2.4205		−2.3946		−2.4337		−2.4512		−2.5088		−2.4566	
R Squared	0.1469	0.5322	0.1480	0.5325	0.1455	0.5324	0.1449	0.5323	0.1425	0.5327	0.1447	0.5332

Notes: FMOLS denotes Fully Modified Least Squares, DOLS denotes Dynamic Ordinary Least Squares, BIND represents Board independence, ROA means Return on Assets, and BIG4 is big four audit companies. ** Significance at 5% level. *** Significance at 1% level.

Table 10. FMOLS and DOLS regression results for the relationship between board characteristics and environmental CSR performance.

Variables	Column (I)		Column (II)		Column (III)		Column (IV)		Column (V)		Column (VI)	
	FMOLS	DOLS	FMOLS	DOLS	FMOLS	DOLS	FMOLS	DOLS	FMOLS	DOLS	FMOLS	DOLS
Board Size	0.0053 ***	0.00021 ***									0.0061 ***	0.0009 ***
BIND			0.0125 ***	0.0020 ***							0.0015 ***	0.0059 ***
CEO duality					−0.0020 ***	−0.0382 ***					−0.0021 ***	−0.0383 ***
Board Mtg							0.0033 ***	0.0068 ***			0.0039 ***	0.0007 ***
Board C'ttee									0.1363 ***	0.013 ***	0.0177 ***	0.0137 ***
Firm Size	0.2208 ***	0.1421 ***	0.2203 **	0.142 ***	0.2139 ***	0.1426 ***	0.2186 ***	0.1431 ***	0.212 ***	0.1405 ***	0.2243 ***	0.1424 ***
Leverage	2.49×10^{-7}	-1.93×10^{-5}	-3.52×10^{-7}	-1.93×10^{-5}	-2.03×10^{-7}	-1.91×10^{-5}	-1.16×10^{-6}	-1.92×10^{-5}	-1.08×10^{-6}	-1.90×10^{-5}	-2.17×10^{-6}	-1.88×10^{-5}
ROA	0.0080 ***	0.0024 **	0.0080 **	0.0024 ***	0.0075 ***	0.00232 ***	0.0079 ***	0.0024 ***	0.00787 ***	0.0023 ***	0.0083 *	0.0023 ***
Firm Age	0.0236 ***	0.644 **	0.0237 **	0.0644 ***	0.0239 ***	0.0645 ***	0.024 ***	0.064 ***	0.0235 ***	0.064 ***	0.0234 ***	0.0642 ***
BIG4	−0.0113	−0.0195	−0.0121	−0.0195	−0.01152	−0.0210	−0.0127	−0.0196	−0.0064	0.0639	−0.0072	−0.0221
Obs	9842	9842	9842	9842	9842	9842	9842	9842	9842	9842	9842	9842
Constant	−1.906		−1.909		−1.892		−1.908		−1.914		−1.954	
R Squared	0.136	0.571	0.1369	0.571	0.1349	0.571	0.1363	0.5711	0.1382	0.5712	0.140	0.572

Notes: FMOLS denotes Fully Modified Least Squares, DOLS denotes Dynamic Ordinary Least Squares, BIND represents Board independence, ROA means Return on Assets, and BIG4 is big four audit companies. * Significance at 10% level. ** Significance at 5% level. *** Significance at 1% level.

Table 11. FMOLS and DOLS regression results for the relationship between board characteristics and social CSR performance.

Variables	Column (I)		Column (II)		Column (III)		Column (IV)		Column (V)		Column (VI)	
	FMOLS	DOLS	FMOLS	DOLS	FMOLS	DOLS	FMOLS	DOLS	FMOLS	DOLS	FMOLS	DOLS
Board Size	0.0032 ***	0.00355 ***									0.0011 ***	0.0067 ***
BIND			0.009 ***	0.0004 ***							0.0075 ***	0.014 ***
CEO Duality					−0.030 ***	−0.026 ***					−0.029 ***	−0.0248 ***
Board Mtg							0.0006 ***	0.0001 ***			0.0011 ***	4.31×10^{-5} ***
Board C'ttee									0.0189 ***	0.0176 ***	0.02033 **	0.018 ***
Firm Size	0.203	0.1581	0.2034	0.1591	0.2004	0.1594	0.1999	0.1589	0.1951	0.1569	0.2034	0.156
Leverage	-7.79×10^{-6}	-3.31×10^{-6}	-7.94×10^{-6}	-3.29×10^{-6}	-6.59×10^{-6}	-3.43×10^{-6}	-7.69×10^{-6}	-3.27×10^{-6}	-8.66×10^{-6}	-3.62×10^{-6}	-7.33×10^{-6}	-3.57×10^{-6}
ROA	0.0065	0.0015	0.0065	0.0016	0.0064	0.00153	0.0064	0.0016	0.0064	0.0015	0.0067	0.0015
Firm Age	0.025	0.0638	0.0249	0.0636	0.025	0.0637	0.0251	0.0636	0.025	0.0631	0.025	0.0634
BIG4	−0.0181	−0.057	−0.01867	−0.059	−0.0178	−0.0596	−0.0181	−0.057	−0.0116	−0.0599	−0.0126	−0.061
Obs	9842	9842	9842	9842	9842	9842	9842	9842	9842	9842	9842	9842
Constant	−1.771		−1.775		−1.780		−1.766		−1.792		−1.834	
R Squared	0.094	0.582	0.095	0.583	0.0919	0.583	0.0925	0.5824	0.0969	0.583	0.099	0.583

Notes: FMOLS denotes Fully Modified Least Squares, DOLS denotes Dynamic Ordinary Least Squares, BIND represents Board independence, ROA means Return on Assets, and BIG4 is big four audit companies. ** Significance at 5% level. *** Significance at 1% level.

In column II, the second variable, board independence (BIND), is positive and significant in Tables 9–11. The results indicate that a percentage point increase in board independence increases the chances of information disclosure on economic CSR (in Table 9), environmental CSR (in Table 10), and social CSR (in Table 11). Therefore, these results suggest that the board's autonomy has the potential to improve information sharing on sustainability. In this sense, an independent board is one that can present information without interference or intervention. This is articulated well by Mahmood et al. (2018) [15], who note that an independent board is not just absence of financial ties with the administration, but also includes a clear presentation of information regarding the administration of the company with a high degree of accuracy. As a result, an independent board of directors will work to prioritize the well-being of societies that are indirectly impacted by the company's operations. In a similar study, Jizi et al. (2014) [52] found a positive relationship between independent board directors and CSR disclosure in firms in the U.S.

On the other hand, as in Table 8, in column III, the results on CEO duality show a negative correlation with economic, environmental, and social sustainability reporting. That is, a one percent increase in CEO duality reduces economic, environmental, and social CSR, respectively. The results suggest that CEO duality impedes sustainability reporting. CEO duality is when the CEO, besides running the organization at the highest level, also holds the position of chairman of the board. In these situations, the CEO has a great deal of executive ability to quickly make crucial business choices. The CEO's position as board chairman, on the other hand, could cause a conflict of interest, an abuse of power, and a loss of trust in the committee's work and reports on social, environmental, and economic sustainability. Our findings confirm the result of Husted and de Sousa-Filho (2019) [13], who show that CEO duality is negatively related to ESG disclosures. However, the results are inconsistent with those of Jizi et al. (2014) [52], who noticed that CEO duality positively impacts CSR disclosures.

In columns IV, board meetings correlate positively with economic, environmental, and social sustainability performance and reporting. Thus, a one percent increase in board meetings, i.e., attendance, leads to a rise in economic, environmental, and social CSR performance. The results suggest that the board's ability to address stakeholders' concerns regarding economic, environmental, and social issues depends on how frequently the board meets to make those critical decisions. Thus, the frequency at which board meetings are held increases the likelihood of reporting on sustainable information to stakeholders. The findings from this study confirm studies conducted in the US where Jizi (2017) and Jizi et al. (2014) [19,52] observed that board meeting frequency rates significantly and positively impact CSR disclosures.

We also observed a positive relationship between board committees and the three pillars of CSR in column V. It suggests that board committees enhance economic, social, and environmental corporate sustainability reporting. Finally, as in section I, we jointly investigate all the independent variables in column VI. The results in columns I–V remain steady with those estimated in column VI. For the control variables, the results suggest that company size, ROA, and age are positively and statistically significant to the economic and environmental, but insignificant to the social. In contrast, leverage and BIG4 are negative and insignificant to all three dimensions of CSR performance.

7. Conclusions, Policy Implications and Suggestions for Future Studies

This study examines the impact of board characteristics (such as board size, board independence, CEO duality, board meetings, and board committees) on corporate sustainability reporting. Unlike previous studies, we examine this relationship considering the three CSR pillars: economic, environmental, and social, using a balanced panel of companies listed on the Shanghai and Shenzhen Stock Exchanges from 2006 to 2019. More importantly, we made an index that combines the social, economic, and environmental pillars of CSR into one. Previous studies have focused on one or two pillars of CSR, primarily economic or social sustainability reporting, which may lead to erroneous conclusions about how

board characteristics and CSR are linked. Also, unlike other studies, we used both the Fully Modified Ordinary Least Squares (FMOLS) and the Dynamic Ordinary Least Squares (DOLS) estimators to deal with the issue of endogeneity. The outcomes of the analysis are as follows:

- The findings of the panel FMOLS and DOLS estimators indicate a positive and statistically significant relationship between the board size and corporate sustainability reporting. Additionally, for the three dimensions of corporate sustainability reporting, the findings indicate that a larger board size increases economic, environmental, and social corporate sustainability reporting. This indicates that larger boards increase the likelihood of disclosing information regarding economic, environmental, and social sustainability.
- Additionally, we discovered a positive relationship between board independence and corporate sustainability reporting. Similar outcomes are observed for all three dimensions of corporate sustainability reporting. One explanation for the positive relationship is that Chinese companies have more external directors on their boards, who are more likely to promote reporting on social, economic, and environmental sustainability information.
- In the case of CEO duality, the result indicates a negative relationship with corporate sustainability reporting. The results for all three dimensions of corporate sustainability reporting are also negative. This suggests that when a CEO has more than one board position, the likelihood of reporting sustainable information about the company's economic, environmental, and social performance decreases. Although only 13.9% of companies in our sample have the same person occupying the CEO and board's chairmanship position, the trend towards CEO duality should not be countenanced.
- Board meeting characteristics also reveal a positive and significant relationship with corporate sustainability reporting and its three dimensions. Frequent board meetings increase the likelihood of reporting sustainable information about their economic, environmental, and social performances to stakeholders.
- Board committees also reveal a positive and significant relationship with corporate sustainability reporting and its three dimensions. This demonstrates that committees tasked with overseeing the operations of a company's boards increase the likelihood of reporting information relating to economic, environmental, and social sustainability.

Based on the findings mentioned above, the study suggests managerial implications for corporate boards, regulators, and practitioners in China and around the world interested in promoting corporate governance and sustainability reporting towards achieving the United Nations' sustainable development agenda by 2030 and beyond.

- In our sample, the board members ranged from three to twenty. A larger board size increases the likelihood of reporting on corporate sustainability reporting. Therefore, businesses are encouraged to consider expanding their boards to improve sustainability reporting. Nonetheless, future research on the optimal board size would be interesting.
- It is strongly suggested that the number of independent directors on boards be increased to guarantee effective advocacy for issues that are stakeholder-driven.
- The board chairmanship and the position of the chief executive officer should be separated. In other words, the positions should not be held by one person.
- The directors are also encouraged to attend all scheduled board meetings to enhance the board's diligence in performing its duties.
- Moreover, boards should create committees to promote corporate sustainability. Therefore, the formation of board committees responsible for sustainability issues should be encouraged.
- As shown in the descriptive statistics, while some Chinese manufacturing industries have achieved very high CSR, economic, environmental, and social scores, exceeding 70%, others have achieved very low levels, falling below 5%. As a result, many low-scoring companies have much room to improve their CSR performance and impacts

and their economic, environmental, and social performance. Companies with high scores should serve as benchmarks and invest in capacity building to improve their performance for the low-scoring companies.

- Finally, these industries are encouraged not to evaluate their sustainability practices isolated from their overall corporate strategy, because the sector is prone to predictable and unpredictable environmental and social challenges and operational risks that ensure the sustainable development of their industries.

While the current study seems to have provided some understanding regarding the relationship between board characteristics and corporate sustainability reporting, some limitations still need to be explored in future studies.

- First, the study did not consider all board characteristics, but the variables used in this study covered most variables used in previous studies. Other board characteristics which could be considered in future research include board diversity, board experience, board ownership, and board compensation.
- Again, the sample size in this study was limited to listed companies on Shanghai and Shenzhen Stock Exchanges. Future studies can be extended to non-listed companies and compare the two categories.
- Finally, the study used a GRI-based index. Future studies can also explore other indices like integrated reporting and triple-bottom approaches.

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